

Gábor Hosu

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I combine a strong mathematical background with hands-on ML engineering – from building RAG pipelines to reproducing ML research papers. I am most interested in designing and training models that solve real problems, and learning how to bring them to production.

Honours and Publications

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| Á. Kránics, G. Hosu , et al., “Multimodal RAG for Context-Aware Chatbots”,
IEEE SISY 23rd [DOI], [Preprint] | Oct 2025 |
| 2nd Prize at 28th Transylvanian Students’ Scientific Conference | May 2025 |

Skills

Languages: Python, SQL (MS SQL)

ML/AI: PyTorch, Transformers, LangChain, Sentence-Transformers, Hugging Face

Data: NumPy, Pandas, Milvus **Backend/Infra:** FastAPI, SQLAlchemy, Docker, Azure, GitLab CI/CD

Other: Git, Linux **Familiar with:** Java, Spring, React, C++

Experience

Software Engineering Internship | Codespring Jul 2024 - Sept 2024

- Designed and implemented an end-to-end backend pipeline for a **[RAG chatbot]**, handling text and image ingestion, vector embedding, and retrieval.
- Developed REST API endpoints with custom authentication and chat, message, and file management.
- Containerized services with Docker and set up a **GitLab CI/CD** pipeline with linting, unit tests, and automatic deployment to Azure.
- Work continued into a BSc thesis and **IEEE publication**.

Fullstack mentorprogram | Codespring Oct 2023 - May 2024

Pre-internship training program covering fullstack development with Java, Spring, and React.

Projects

Embedding-Based Retrieval Paper Reproduction Nov 2025 - Jan 2026

[Portfolio] | [GitHub] | [Paper] | [HF (Models & Data)]

- Reproduced key results of a **2025 Google DeepMind paper** on geometric limits of single-vector embedding retrieval under GPU constraints.
- Generated synthetic **IR datasets (2K–50K docs)** and quantized models; published artifacts on Hugging Face.
- Fine-tuned the full model pipeline, implementing benchmarking with quantized LLMs and experimenting with datasets and hyperparameters.
- Invited to present findings at a **research group seminar**, recognized beyond the course scope.

Education

Babeş-Bolyai University, MSc, Data Analysis and Modelling Oct 2025 - Present

- Relevant coursework:** Information Retrieval, Swarm Intelligence, Robotics
- Current semester:** Machine Learning, NLP, GPU Programming

Babeş-Bolyai University, BSc, Mathematics and Computer Science 2022 - 2025

GPA: 9.58/10 | Thesis: "A file-centric chatbot based on the RAG method"

Languages

English: Upper-Intermediate

Romanian: Intermediate

Hungarian: Native